

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): LW401, LW551i, LWU421, LWU501i, LX501, LX601i

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.06.0000-b010	2.01.0002-b005	1.4.0

Version History

Module Version	Date	Notes
1_0_2_0	7/7/2007	Added My Memory Load and My Memory Save commands. Fixed Pin Assignments Diagram. Updated driver to latest timing and logics.
1_0_1_1	6/28/2016	Updated Module logic. Fixed Volume Step command.
1_0_1_0	2/11/2016	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'.
- ConnectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.ConnectionCounter = 5.
- If Login Credentials are required, devicePassword variable need to be set to the password use for the device. Example: InterfaceName.Password = 'Admin'.

Control Commands, States, and Qualifiers:

AspectRatio	'Normal' '16:10'	'4:3' '14:9'	'16:9' 'Native' ¹
AudioMute	'On'	'Off'	
AutoEco	'On'	'Off'	
AutoImage	'None'		
ClosedCaption	'On'	'Off'	'Auto'
ClosedCaptionChannel	'1' – '4'		
ClosedCaptionMode	'Captions'	'Text'	
Freeze	'On'	'Off'	
Input	'Computer 1' 'USB Type A' 'HDMI 2' 'Video'	'Computer 2' 'USB Type B' 'Component'	'LAN' 'HDMI 1' 'S-Video'
LampMode	'Normal'	'Eco'	
MyMemoryLoad	'1' – '4'		
MyMemorySave	'1' – '4'		
PIPLLeftSource ¹	'Computer 1' 'HDMI 2' 'Video'	'Computer 2' 'Component'	'HDMI 1' 'S-Video'
PIPMode ¹	'On'	'Off'	
PIPPosition ¹	'Left'	'Right'	
PIPRightSource ¹	'Computer 1' 'HDMI 2' 'Video'	'Computer 2' 'Component'	'HDMI 1' 'S-Video'
PIPSwap ¹	'None'		
Power	'On'	'Off'	
VideoMute	'On'	'Off'	
VolumeStep	'Increment'	'Decrement'	
['InputType']	'Computer 1' 'USB Type A' 'HDMI 2' 'Video'	'Computer 2' 'USB Type B' 'Component'	'LAN' 'HDMI 1' 'S-Video'

¹ Found on all models except LX501 & LX601i

Status Available:

AspectRatio	'Normal' '16:10'	'4:3' '14:9'	'16:9' 'Native' ¹
AudioMute	'On'	'Off'	
AutoEco	'On'	'Off'	
ClosedCaption	'On'	'Off'	'Auto'
ClosedCaptionChannel	'1' – '4'		
ClosedCaptionMode	'Captions'	'Text'	
ConnectionStatus	'Connected'	'Disconnected'	
DeviceStatus	'Normal' 'Lamp Error' 'Cold Error'	'Cover Error' 'Temp Error' 'Filter Error'	'Fan Error' 'Air Flow'
FilterUsage	Hours		
Freeze	'On'	'Off'	
Input	'Computer 1' 'USB Type A' 'HDMI 2' 'Video'	'Computer 2' 'USB Type B' 'Component'	'LAN' 'HDMI 1' 'S-Video'
LampMode	'Normal'	'Eco'	
LampUsage	Hours		
PIPLetSource ¹	'Computer 1' 'HDMI 2' 'Video'	'Computer 2' 'Component'	'HDMI 1' 'S-Video'
PIPMode ¹	'On'	'Off'	
PIPPosition ¹	'Left'	'Right'	
PIPRightSource ¹	'Computer 1' 'HDMI 2' 'Video'	'Computer 2' 'Component'	'HDMI 1' 'S-Video'
Power	'On'	'Off'	'Cooling Down'
VideoMute	'On'	'Off'	
VolumeLevelStatus	0 – 48		
['InputType']	'Computer 1' 'USB Type A' 'HDMI 2' 'Video'	'Computer 2' 'USB Type B' 'Component'	'LAN' 'HDMI 1' 'S-Video'

¹ Found on all models except LX501 & LX601i

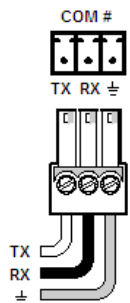
Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

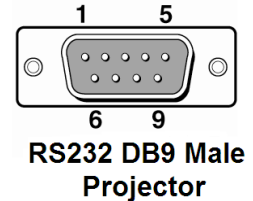
Notes for the Device:

In order to control the device, the following must be set through the device's OSD menu.

- Standby Mode must be set to Normal
- Network Bridge must be set to Off

Serial communication:**Port Type:** RS-232**Baud Rate:** 19200**Data Bits:** 8**Parity:** None**Stop Bits:** One**Flow Control:** None**Pin Assignments Diagram**

Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND

**General Notes:**

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Aspect Ratio 14:9	\xBE\xEF\x03\x06\x00\xCE\xD6\x01\x00\x08 \x09\x00
Aspect Ratio 16:10	\xBE\xEF\x03\x06\x00>\xD6\x01\x00\x08 \x0A\x00
Aspect Ratio 16:9	\xBE\xEF\x03\x06\x00\x0E\xD1\x01\x00\x08 \x01\x00
Aspect Ratio 4:3	\xBE\xEF\x03\x06\x00\x9E\xD0\x01\x00\x08 \x00\x00
Aspect Ratio Native	\xBE\xEF\x03\x06\x00^\xD7\x01\x00\x08 \x08\x00
Aspect Ratio Normal	\xBE\xEF\x03\x06\x00^\xDD\x01\x00\x08 \x10\x00
Audio Mute Off	\xBE\xEF\x03\x06\x00F\xD3\x01\x00\x02 \x00\x00
Audio Mute On	\xBE\xEF\x03\x06\x00\xD6\xD2\x01\x00\x02 \x01\x00
Auto Eco Off	\xBE\xEF\x03\x06\x00\xFB\x01\x00\x103\x00\x00
Auto Eco On	\xBE\xEF\x03\x06\x00k&\x01\x00\x103\x01\x00
Auto Image None	\xBE\xEF\x03\x06\x00\x91\xD0\x06\x00\x0A \x00\x00
Closed Caption Auto	\xBE\xEF\x03\x06\x00\x9A\x01\x00\x007\x02\x00
Closed Caption Channel 1	\xBE\xEF\x03\x06\x00\xD2b\x01\x00\x027\x01\x00
Closed Caption Channel 4	\xBE\xEF\x03\x06\x00\x82a\x01\x00\x027\x04\x00
Closed Caption Mode Captions	\xBE\xEF\x03\x06\x00\x06c\x01\x00\x017\x00\x00
Closed Caption Mode Text	\xBE\xEF\x03\x06\x00\x96b\x01\x00\x017\x01\x00
Closed Caption Off	\xBE\xEF\x03\x06\x00\xFAb\x01\x00\x007\x00\x00
Closed Caption On	\xBE\xEF\x03\x06\x00jc\x01\x00\x007\x01\x00
Freeze Off	\xBE\xEF\x03\x06\x00\x83\xD2\x01\x00\x020\x00\x00
Freeze On	\xBE\xEF\x03\x06\x00\x13\xD3\x01\x00\x020\x01\x00
Input Component	\xBE\xEF\x03\x06\x00\xAE\xD1\x01\x00\x00 \x05\x00
Input Computer 1	\xBE\xEF\x03\x06\x00\xFE\xD2\x01\x00\x00 \x00\x00
Input Computer 2	\xBE\xEF\x03\x06\x00>\xD0\x01\x00\x00 \x04\x00
Input HDMI 1	\xBE\xEF\x03\x06\x00\x0E\xD2\x01\x00\x00 \x03\x00
Input HDMI 2	\xBE\xEF\x03\x06\x00n\xD6\x01\x00\x00 \x0D\x00
Input LAN	\xBE\xEF\x03\x06\x00\xCE\xD5\x01\x00\x00 \x0B\x00
Input S-Video	\xBE\xEF\x03\x06\x00\x9E\xD3\x01\x00\x00 \x02\x00
Input USB Type A	\xBE\xEF\x03\x06\x00^\xD1\x01\x00\x00 \x06\x00
Input USB Type B	\xBE\xEF\x03\x06\x00\xFE\xD7\x01\x00\x00 \x0C\x00
Input Video	\xBE\xEF\x03\x06\x00n\xD3\x01\x00\x00 \x01\x00
Lamp Mode Eco	\xBE\xEF\x03\x06\x00\xAB"\x01\x00\x003\x01\x00
Lamp Mode Normal	\xBE\xEF\x03\x06\x00;\x01\x00\x003\x00\x00
My Memory Load 1	\xBE\xEF\x03\x06\x00\x0E\xD7\x01\x00\x14 \x00\x00
My Memory Load 4	\xBE\xEF\x03\x06\x00\xFE\xD7\x01\x00\x14 \x03\x00
My Memory Save 1	\xBE\xEF\x03\x06\x00\xF2\xD6\x01\x00\x15 \x00\x00
My Memory Save 4	\xBE\xEF\x03\x06\x00\x02\xD6\x01\x00\x15 \x03\x00
PIP Left Source Component	\xBE\xEF\x03\x06\x00xA2%\x01\x00\x15#\x05\x00
PIP Left Source Computer 1	\xBE\xEF\x03\x06\x00\xF2&\x01\x00\x15#\x00\x00
PIP Left Source Computer 2	\xBE\xEF\x03\x06\x002\$\x01\x00\x15#\x04\x00
PIP Left Source HDMI 1	\xBE\xEF\x03\x06\x00\x02&\x01\x00\x15#\x03\x00
PIP Left Source HDMI 2	\xBE\xEF\x03\x06\x00b"\x01\x00\x15#\x0D\x00
PIP Left Source S-Video	\xBE\xEF\x03\x06\x00\x92"\x01\x00\x15#\x02\x00
PIP Left Source Video	\xBE\xEF\x03\x06\x00b"\x01\x00\x15#\x01\x00
PIP Mode Off	\xBE\xEF\x03\x06\x00>&\x01\x00\x10#\x00\x00
PIP Mode On	\xBE\xEF\x03\x06\x00\xAE"\x01\x00\x10#\x01\x00
PIP Position Left	\xBE\xEF\x03\x06\x00z&\x01\x00\x13#\x00\x00

PIP Position Right	\xBE\xEF\x03\x06\x00\xEA\x01\x00\x13#\x01\x00
PIP Right Source Component	\xBE\xEF\x03\x06\x00\xD6\$\x01\x00\x12#\x05\x00
PIP Right Source Computer 1	\xBE\xEF\x03\x06\x00\x86'\x01\x00\x12#\x00\x00
PIP Right Source Computer 2	\xBE\xEF\x03\x06\x00F%\x01\x00\x12#\x04\x00
PIP Right Source HDMI 1	\xBE\xEF\x03\x06\x00v'\x01\x00\x12#\x03\x00
PIP Right Source HDMI 2	\xBE\xEF\x03\x06\x00\x16#\x01\x00\x12#\x0D\x00
PIP Right Source S-Video	\xBE\xEF\x03\x06\x00\xE6&\x01\x00\x12#\x02\x00
PIP Right Source Video	\xBE\xEF\x03\x06\x00\x16&\x01\x00\x12#\x01\x00
PIP Swap None	\xBE\xEF\x03\x06\x00\x01'\x06\x00\x16#\x00\x00
Power Off	\xBE\xEF\x03\x06\x00*\xD3\x01\x00\x00\x00\x00
Power On	\xBE\xEF\x03\x06\x00\xBA\xD2\x01\x00\x00\x01\x00
Video Mute Off	\xBE\xEF\x03\x06\x00\xFB\xD8\x01\x00 0\x00\x00
Video Mute On	\xBE\xEF\x03\x06\x00k\xD9\x01\x00 0\x01\x00
Volume Step Decrement Input Type Component	\xBE\xEF\x03\x06\x00\xB6\xCD\x05\x00e \x00\x00
Volume Step Decrement Input Type Computer 1	\xBE\xEF\x03\x06\x00z\xCD\x05\x00` \x00\x00
Volume Step Decrement Input Type Computer 2	\xBE\xEF\x03\x06\x00J\xCC\x05\x00d \x00\x00
Volume Step Decrement Input Type HDMI 1	\xBE\xEF\x03\x06\x00>\xCD\x05\x00c \x00\x00
Volume Step Decrement Input Type HDMI 2	\xBE\xEF\x03\x06\x00\xD6\xCF\x05\x00m \x00\x00
Volume Step Decrement Input Type LAN	\xBE\xEF\x03\x06\x00^\xCF\x05\x00k \x00\x00
Volume Step Decrement Input Type S-Video	\xBE\xEF\x03\x06\x00\xC2\xCC\x05\x00b \x00\x00
Volume Step Decrement Input Type USB Type A	\xBE\xEF\x03\x06\x00\xF2\xCD\x05\x00f \x00\x00
Volume Step Decrement Input Type USB Type B	\xBE\xEF\x03\x06\x00*\xCE\x05\x00l \x00\x00
Volume Step Decrement Input Type Video	\xBE\xEF\x03\x06\x00\x86\xCC\x05\x00a \x00\x00
Volume Step Increment Input Type Component	\xBE\xEF\x03\x06\x00g\xCC\x04\x00e \x00\x00
Volume Step Increment Input Type Computer 1	\xBE\xEF\x03\x06\x00\xAB\xCC\x04\x00` \x00\x00
Volume Step Increment Input Type Computer 2	\xBE\xEF\x03\x06\x00\x9B\xCD\x04\x00d \x00\x00
Volume Step Increment Input Type HDMI 1	\xBE\xEF\x03\x06\x00\xEF\xCC\x04\x00c \x00\x00
Volume Step Increment Input Type HDMI 2	\xBE\xEF\x03\x06\x00\x07\xCE\x04\x00m \x00\x00
Volume Step Increment Input Type LAN	\xBE\xEF\x03\x06\x00\x8F\xCE\x04\x00k \x00\x00
Volume Step Increment Input Type S-Video	\xBE\xEF\x03\x06\x00\x13\xCD\x04\x00b \x00\x00
Volume Step Increment Input Type USB Type A	\xBE\xEF\x03\x06\x00#\xCC\x04\x00f \x00\x00
Volume Step Increment Input Type USB Type B	\xBE\xEF\x03\x06\x00\xFB\xCF\x04\x00l \x00\x00
Volume Step Increment Input Type Video	\xBE\xEF\x03\x06\x00W\xCD\x04\x00a \x00\x00

Appendix B. Query Commands

Aspect Ratio	\xBE\xEF\x03\x06\x00\xAD\xD0\x02\x00\x08 \x00\x00
Audio Mute	\xBE\xEF\x03\x06\x00u\xD3\x02\x00\x02 \x00\x00
Auto Eco	\xBE\xEF\x03\x06\x00\xC8'\x02\x00\x103\x00\x00
Closed Caption	\xBE\xEF\x03\x06\x00\xC9b'\x02\x00\x007\x00\x00
Closed Caption Channel	\xBE\xEF\x03\x06\x00qc'\x02\x00\x027\x00\x00
Closed Caption Mode	\xBE\xEF\x03\x06\x005c'\x02\x00\x017\x00\x00
Device Status	\xBE\xEF\x03\x06\x00\xD9\xD8\x02\x00 \x00\x00
Filter Usage	\xBE\xEF\x03\x06\x00\xC2\xF0\x02\x00\xA0\x10\x00\x00
Freeze	\xBE\xEF\x03\x06\x00\xB0\xD2\x02\x00\x020\x00\x00
Input	\xBE\xEF\x03\x06\x00\xCD\xD2\x02\x00\x00 \x00\x00
Lamp Mode	\xBE\xEF\x03\x06\x00\x08#\x02\x00\x003\x00\x00
Lamp Usage	\xBE\xEF\x03\x06\x00\xC2\xFF\x02\x00\x90\x10\x00\x00

**Global Scripter Module
Communication Sheet**

PIP Left Source	\xBE\xEF\x03\x06\x00\xC1&\x02\x00\x15#\x00\x00
PIP Mode	\xBE\xEF\x03\x06\x00\x0D&\x02\x00\x10#\x00\x00
PIP Position	\xBE\xEF\x03\x06\x00I&\x02\x00\x13#\x00\x00
PIP Right Source	\xBE\xEF\x03\x06\x00\xB5'\x02\x00\x12#\x00\x00
Power	\xBE\xEF\x03\x06\x00\x19\xD3\x02\x00\x00`\x00\x00
Video Mute	\xBE\xEF\x03\x06\x00\xC8\xD8\x02\x00 0\x00\x00
Volume Level Status Input Type Component	\xBE\xEF\x03\x06\x00\x01\xCC\x02\x00e \x00\x00
Volume Level Status Input Type Computer 1	\xBE\xEF\x03\x06\x00\xCD\xCC\x02\x00` \x00\x00
Volume Level Status Input Type Computer 2	\xBE\xEF\x03\x06\x00\xFD\xCD\x02\x00d \x00\x00
Volume Level Status Input Type HDMI 1	\xBE\xEF\x03\x06\x00\x89\xCC\x02\x00c \x00\x00
Volume Level Status Input Type HDMI 2	\xBE\xEF\x03\x06\x00a\xCE\x02\x00m \x00\x00
Volume Level Status Input Type LAN	\xBE\xEF\x03\x06\x00\xE9\xCE\x02\x00k \x00\x00
Volume Level Status Input Type S-Video	\xBE\xEF\x03\x06\x00u\xCD\x02\x00b \x00\x00
Volume Level Status Input Type USB Type A	\xBE\xEF\x03\x06\x00E\xCC\x02\x00f \x00\x00
Volume Level Status Input Type USB Type B	\xBE\xEF\x03\x06\x00\x9D\xCF\x02\x00l \x00\x00
Volume Level Status Input Type Video	\xBE\xEF\x03\x06\x001\xCD\x02\x00a \x00\x00